

Lab 1:

Query

Query History

1

SELECT

2

relname AS object_name,

3

pg_size_pretty (pg_table_size (fed . oid)) AS data_size ,

4

pg_size_pretty (pg_indexes_size (fed . oid)) AS index_size ,

5

pg_size_pretty(pg_total_relation_size(fed.oid)) AS total_size FROM pg_class fed

6

JOIN pg_namespace n ON n.oid = fed.relnamespace

7

WHERE fed. relkind = 'r' AND n.nspname = 'public'

8

ORDER BY pg_total_relation_size (fed . oid) DESC ;

Data Output

Messages

Notifications

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In this database, table rental and payment contain much more index_size than the others because of their structure and business activity.

+ **High Business & Transaction Volume:** As the core transactional (fact) tables of the rental business, rental and payment log every user activity and financial transaction.

+ **Heavy Indexing on Foreign Keys & Constraints:** To support frequent relational joins and maintain data integrity, both tables maintain multiple B-Tree indexes

Lab 2:

Query

Query History

1

`SELECT pg_size_pretty(pg_indexes_size('payment'));`

2

`CREATE INDEX idx_payment_date_long ON payment (payment_date, amount, customer_id );`

3

`SELECT pg_size_pretty(pg_indexes_size('payment'));`

Data Output

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Showing rows: 1 to 1

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	pg_size_pretty text
1	1488 kB

Query

Query History

1

`SELECT`

2

`relname AS index_name,`

3

`relpages AS page_count,`

4

`pg_size_pretty(relpages * 8192::bigint) AS total_size`

5

`FROM pg_class`

6

`WHERE relname = 'idx_payment_date_long';`

Data Output

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	index_name name	page_count integer	total_size text
1	idx_payment_date_lo...	71	568 kB

The newly created index `idx_payment_date_long` consumed **71 Pages** (71×8 KB=568 KB).

Lab 3:

Query

Query History

1

CREATE INDEX idx_expensive_films ON film (replacement_cost)

2

WHERE replacement_cost > 25.00;

3

4

CREATE INDEX idx_all_films_cost ON film (replacement_cost);

5

6

SELECT

7

relname AS index_name,

8

relpages AS page_count,

9

pg_size_pretty(pg_relation_size(oid)) AS index_size

10

FROM pg_class

11

WHERE relname IN ('idx_expensive_films', 'idx_all_films_cost');

Data Output

Messages

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	index_name name	page_count integer	index_size text
1	idx_expensive_fil...	2	16 kB
2	idx_all_films_cost	5	40 kB